

Red Hat OpenShift Data Foundation 4.22

Updating OpenShift Data Foundation

Instructions for cluster and storage administrators regarding upgrading

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Red Hat OpenShift Data Foundation 4.22 Updating OpenShift Data Foundation

Instructions for cluster and storage administrators regarding upgrading

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Abstract

This document explains how to update previous versions of Red Hat OpenShift Data Foundation.

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PREFACE

This document describes how to update Red Hat OpenShift Data Foundation and its components to a new version.

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION

We appreciate your input on our documentation. Do let us know how we can make it better.

To give feedback, create a Jira ticket:

Procedure

1. Log in to the Jira.
2. Click **Create** in the top navigation bar
3. Enter a descriptive title in the Summary field.
4. Enter your suggestion for improvement in the Description field. Include links to the relevant parts of the documentation.
5. Select **Documentation** in the Components field.
6. Click Create at the bottom of the dialogue.

CHAPTER 1. OVERVIEW OF THE OPENSHIFT DATA FOUNDATION UPDATE PROCESS

This chapter helps you to upgrade between the minor releases and z-streams for all Red Hat OpenShift Data Foundation deployments (Internal, Internal-Attached and External). The upgrade process remains the same for all deployments.

You can upgrade OpenShift Data Foundation and its components, either between minor releases like 4.21 and 4.22, or between z-stream updates like 4.22.0 and 4.22.1 by enabling automatic updates (if not done so during operator installation) or performing manual updates. When a new z-stream release becomes available, the upgrade process triggers automatically if the update strategy was set to Automatic. It is recommended to take snapshots of all the volumes in use before running any maintenance or upgrade procedures.

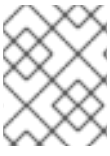
Extended Update Support (EUS)

EUS to EUS upgrade in OpenShift Data Foundation is sequential and it is aligned with OpenShift upgrade. For more information, see *Performing a Control Plane Only update* and *Control Plane Only updates for layered products and Operators installed through Operator Lifecycle Manager* in the OpenShift Container Platform documentation.

For EUS upgrade of OpenShift Container Platform and OpenShift Data Foundation, make sure that OpenShift Data Foundation is upgraded along with OpenShift Container Platform and compatibility between OpenShift Data Foundation and OpenShift Container Platform is always maintained.

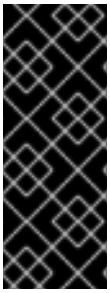
Example workflow of EUS upgrade:

1. Pause the worker machine pools.
2. Update OpenShift <4.y> → OpenShift <4.y+1>.
3. Update OpenShift Data Foundation <4.y> → OpenShift Data Foundation <4.y+1>.
4. Update OpenShift <4.y+1> → OpenShift <4.y+2>.
5. Update to OpenShift Data Foundation <4.y+2>.
6. Unpause the worker machine pools.



NOTE

You can update to ODF <4.y+2> either before or after worker machine pools are unpaused.



IMPORTANT

When you update OpenShift Data Foundation in external mode, make sure that the Red Had Ceph Storage and OpenShift Data Foundation versions are compatible. For more information about supported Red Had Ceph Storage version in external mode, refer to the *Red Hat OpenShift Data Foundation Supportability and Interoperability Checker* . Provide the required OpenShift Data Foundation version in the checker to see the supported Red Had Ceph version corresponding to the version in use.

You also need to upgrade the different parts of Red Hat OpenShift Data Foundation in the following order for both internal and external mode deployments:

1. **Update OpenShift Container Platform** according to the *Updating clusters* documentation for OpenShift Container Platform.

2. **Update Red Hat OpenShift Data Foundation.**

    a. **To prepare a disconnected environment for updates**, see *Using Operator Lifecycle Manager in disconnected environments* to be able to update OpenShift Data Foundation as well as Local Storage Operator when in use. For the list of packages to be mirrored for upgrading OpenShift Data Foundation in disconnected environments, see the Disconnected environment section in the *Planning your deployment* guide.

    b. **For updating between minor releases**, see *Updating Red Hat OpenShift Data Foundation 4.20 to 4.22*.

    c. **For updating between z-stream releases**, see *Updating Red Hat OpenShift Data Foundation 4.22.x to 4.22.y*.

    d. **For updating external mode deployments**, you must also perform the steps from the *Updating the Red Hat OpenShift Data Foundation external secret* section.

    e. **If you use local storage, then update the Local Storage operator**See *Checking for Local Storage Operator deployments* if you are unsure.



IMPORTANT

If you have an existing setup of OpenShift Data Foundation 4.21 with disaster recovery (DR) enabled, ensure to update all your clusters in the environment at the same time and avoid updating a single cluster. This is to avoid any potential issues and maintain best compatibility. It is also important to maintain consistency across all OpenShift Data Foundation DR instances.

Update considerations

Review the following important considerations before you begin.

- The Red Hat OpenShift Container Platform version is the same as Red Hat OpenShift Data Foundation. See the *Interoperability Matrix* for more information about supported combinations of OpenShift Container Platform and Red Hat OpenShift Data Foundation.

- To know whether your cluster was deployed in internal or external mode, see *How to determine if ODF cluster has storage in internal or external mode*.

- The Local Storage Operator is fully supported only when the Local Storage Operator version matches the Red Hat OpenShift Container Platform version.

- In OpenShift Data Foundation clusters with disaster recovery (DR) enabled, during upgrade to version 4.22, previous **bluestore-rdr** OSDs are migrated to **bluestore** OSDs. **bluestore** backed OSDs now provide the same improved performance of **bluestore-rdr** based OSDs, which is important when the cluster is required to be used for Regional Disaster Recovery.
  During upgrade you can view the status of the OSD migration. In the OpenShift Web Console, navigate to **Storage → Data Foundation → Storage System**. In the **Activity** card of the **Block and File** tab you can view ongoing activities. **Migrating cluster OSDs** shows the status of the migration from **bluestore-rdr** to **bluestore**.



IMPORTANT

To avoid data loss, it is recommended to take a backup of NooBaa DB PVC regularly. If NooBaa DB fails and cannot be recovered, then you can revert to the latest backed-up version. For instructions on backing up NooBaa DB, follow the steps in *Perform a One-Time Backup of the Database for the Multicloud Object Gateway*.

Additional resources

- Performing a Control Plane Only update
- Control Plane Only updates for layered products and Operators installed through Operator Lifecycle Manager
- Red Hat OpenShift Data Foundation Supportability and Interoperability Checker
- Updating clusters
- Using Operator Lifecycle Manager in disconnected environments
- Disconnected environment
- Updating Red Hat OpenShift Data Foundation 4.20 to 4.22
- Updating Red Hat OpenShift Data Foundation 4.22.x to 4.22.y
- Updating the Red Hat OpenShift Data Foundation external secret
- Checking for Local Storage Operator deployments
- Interoperability Matrix
- How to determine if ODF cluster has storage in internal or external mode
- Perform a One-Time Backup of the Database for the Multicloud Object Gateway

CHAPTER 2. OPENSHIFT DATA FOUNDATION UPGRADE CHANNELS AND RELEASES

In OpenShift Container Platform 4.1, Red Hat introduced the concept of channels for recommending the appropriate release versions for cluster upgrades. By controlling the pace of upgrades, these upgrade channels allow you to choose an upgrade strategy. As OpenShift Data Foundation gets deployed as an operator in OpenShift Container Platform, it follows the same strategy to control the pace of upgrades by shipping the fixes in multiple channels. Upgrade channels are tied to a minor version of OpenShift Data Foundation.

For example, OpenShift Data Foundation 4.22 upgrade channels recommend upgrades within 4.22. Upgrades to future releases is not recommended. This strategy ensures that administrators can explicitly decide to upgrade to the next minor version of OpenShift Data Foundation.

Upgrade channels control only release selection and do not impact the version of the cluster that you install; the **odf-operator** decides the version of OpenShift Data Foundation to be installed. By default, it always installs the latest OpenShift Data Foundation release maintaining the compatibility with OpenShift Container Platform. So, on OpenShift Container Platform 4.22, OpenShift Data Foundation 4.22 will be the latest version which can be installed.

OpenShift Data Foundation upgrades are tied to the OpenShift Container Platform upgrade to ensure that compatibility and interoperability are maintained with the OpenShift Container Platform. For OpenShift Data Foundation 4.22, OpenShift Container Platform 4.22 and 4.23 (when generally available) are supported. OpenShift Container Platform 4.22 is supported to maintain forward compatibility of OpenShift Data Foundation with OpenShift Container Platform. Keep the OpenShift Data Foundation version the same as OpenShift Container Platform in order to get the benefit of all the features and enhancements in that release.



IMPORTANT

Due to fundamental Kubernetes design, all OpenShift Container Platform updates between minor versions must be serialized. You must update from OpenShift Container Platform 4.20 to 4.21 and then to 4.22. You cannot update from OpenShift Container Platform 4.20 to 4.22 directly. For more information, see *Preparing to perform an EUS-to-EUS update* in the *Updating clusters* guide in the OpenShift Container Platform documentation.

OpenShift Data Foundation 4.22 offers the following upgrade channel:

- stable-4.22
- stable-4.21

stable-4.22 channel

Once a new version is Generally Available, the stable channel corresponding to the minor version gets updated with the new image which can be used to upgrade. You can use the **stable-4.22** channel to upgrade from OpenShift Data Foundation 4.21 and upgrades within 4.22.

stable-4.21

You can use the **stable-4.21** channel to upgrade from OpenShift Data Foundation 4.20 and upgrades within 4.21.

CHAPTER 3. UPDATING RED HAT OPENSHIFT DATA FOUNDATION 4.21 TO 4.22

This chapter helps you to upgrade between the minor releases for all Red Hat OpenShift Data Foundation deployments (Internal, Internal-Attached and External). The upgrade process remains the same for all deployments. The Only difference is what gets upgraded and what's not.

- For Internal and Internal-attached deployments, upgrading OpenShift Data Foundation upgrades all OpenShift Data Foundation services including the backend Red Hat Ceph Storage (RHCS) cluster.
- For External mode deployments, upgrading OpenShift Data Foundation only upgrades the OpenShift Data Foundation service while the backend Ceph storage cluster remains untouched and needs to be upgraded separately.
  You must upgrade Red Hat Ceph Storage along with OpenShift Data Foundation to get new feature support, security fixes, and other bug fixes. As there is no dependency on RHCS upgrade, you can upgrade the OpenShift Data Foundation operator first followed by RHCS upgrade or vice-versa. For more information about RHCS releases, see the knowledgebase solution, Red Hat Ceph Storage releases and corresponding Ceph package versions .



IMPORTANT

Upgrading to 4.22 directly from any version older than 4.21 is not supported.

When upgrading from OpenShift Data Foundation 4.21.z to 4.22, you must upgrade to OpenShift Data Foundation 4.22.2 or later versions. Upgrades to OpenShift Data Foundation 4.22.0 or 4.22.1 are not supported and will fail.

Prerequisites

- Ensure that the **OpenShift Container Platform** cluster has been updated to the latest stable release of version 4.22.x, see Updating Clusters.
- Ensure that the **OpenShift Data Foundation** cluster is healthy and data is resilient.
  - Navigate to **Storage → Data Foundation → Storage Systems** tab and then click on the storage system name.
  - Check for the green tick on the status card of both **Overview - Block and File** and **Object** tabs. Green tick indicates that the *storage cluster*, *object service* and *data resiliency* are all healthy.
- Ensure that all **OpenShift Data Foundation** Pods, including the operator pods, are in **Running** state in the **openshift-storage** namespace.
  To view the state of the pods, on the OpenShift Web Console, click **Workloads → Pods**. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Ensure that you have sufficient time to complete the OpenShift Data Foundation update process, as the update time varies depending on the number of OSDs that run in the cluster.

- Optional: To reduce the upgrade time for large clusters, it is encouraged to increase the **maxUnavailable** for **NodePlugin**. Use the following command:

```
VAL=$(oc get nodes -o name | awk 'END {printf "%.0f\n", NR * 0.2}') && oc patch driver openshift-storage.rbd.csi.ceph.com -n openshift-storage --type='merge' -p "{\"spec\":{\"nodePlugin\":{\"updateStrategy\":{\"rollingUpdate\":{\"maxUnavailable\":$VAL}}}}}" && oc patch driver openshift-storage.cephfs.csi.ceph.com -n openshift-storage --type='merge' -p "{\"spec\":{\"nodePlugin\":{\"updateStrategy\":{\"rollingUpdate\":{\"maxUnavailable\":$VAL}}}}}"
```

This affects CephFS and CephRBD daemonsets and allows the pods to restart simultaneously or be unavailable and reduce the upgrade time. For an optimal value, you can set the parameter values to 20%. However, if the value is too high, disruption for new volumes might be observed during the upgrade.

- **Prerequisite relevant only for OpenShift Data Foundation deployments on AWS using AWS Security Token Service (STS)**
  Add another entry in the trust policy for **noobaa-core** account as follows:

  1. Log into AWS web console where the AWS role resides using http://console.aws.amazon.com/.
  2. Enter the IAM management tool and click **Roles**.
  3. Find the name of the role created for AWS STS to support Multicloud Object Gateway (MCG) authentication using the following command in OpenShift CLI:

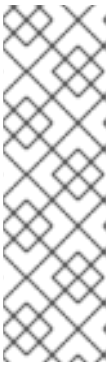
     ```
     $ oc get deployment noobaa-operator -o yaml -n openshift-storage | grep ROLEARN -A1
           value: arn:aws:iam::123456789101:role/your-role-name-here
     ```

  4. Search for the role name that you obtained from the previous step in the tool and click on the role name.
  5. Under the role summary, click **Trust relationships**.
  6. In the **Trusted entities** tab, click **Edit trust policy** on the right.
  7. Under the **"Action": "sts:AssumeRoleWithWebIdentity"** field, there are two fields to enable access for two NooBaa service accounts **noobaa** and **noobaa-endpoint**. Add another entry for the core pod's new service account name, **system:serviceaccount:openshift-storage:noobaa-core**.
  8. Click **Update policy** at the bottom right of the page.
     The update might take about 5 minutes to get in place.

Procedure

1. On the OpenShift Web Console, navigate to **Ecosystem → Installed Operators**.
2. Select **openshift-storage** project.
3. Click the OpenShift Data Foundation operator name.
4. Click the **Subscription** tab and click the link under **Update Channel**.
5. Select the **stable-4.22** update channel and **Save** it.

6. If the **Upgrade status** shows **requires approval**, click on **requires approval**.
   a. On the Install Plan Details page, click **Preview Install Plan**.
   b. Review the install plan and click **Approve**.
   Wait for the **Status** to change from **Unknown** to **Created**.
7. Navigate to **Ecosystem → Installed Operators**.
8. Select the **openshift-storage** project.
   Wait for the OpenShift Data Foundation Operator **Status** to change to **Up to date**.
9. After the operator is successfully upgraded, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.

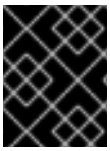


NOTE

After upgrading, if your cluster has five or more nodes, racks, or rooms, and when there are five or more number of failure domains present in the deployment, you can configure Ceph monitor counts based on the number of racks or zones. An alert is displayed in the notification panel or Alert Center of the OpenShift Web Console to indicate the option to increase the number of Ceph monitor counts. You can use the **Configure** option in the alert to configure the Ceph monitor counts. For more information, see *Resolving low Ceph monitor count alert* in the Troubleshooting guide.

Verification

- Check the **Version** after the OpenShift Data Foundation name and check the operator status.
  - Navigate to **Ecosystem → Installed Operators** and select the **openshift-storage** project.
  - When the upgrade completes, the version updates to a new version number for OpenShift Data Foundation and status changes to **Succeeded** with a green tick.
- Verify that the **OpenShift Data Foundation** cluster is healthy and data is resilient.
  - Navigate to **Storage → Data Foundation → Storage Systems** tab and then click on the storage system name.
  - Check for the green tick on the status card of **Overview- Block and File** and **Object** tabs. Green tick indicates that the storage cluster, object service and data resiliency is healthy.
- If verification steps fail, contact Red Hat Support .



IMPORTANT

After updating external mode deployments, you must also update the external secret. For instructions, see *Updating the OpenShift Data Foundation external secret* .

Additional resources

- Updating the OpenShift Data Foundation external secret
- Commonly required logs for troubleshooting

- Resolving low Ceph monitor count alert

CHAPTER 4. UPDATING RED HAT OPENSHIFT DATA FOUNDATION 4.22.X TO 4.22.Y

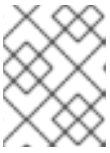
Apply a OpenShift Data Foundation z-stream update within the 4.22 release to receive bug fixes and security patches without changing the minor version, using either the automatic approval strategy or by manually approving the generated InstallPlan.

- For Internal and Internal-attached deployments, upgrading OpenShift Data Foundation upgrades all OpenShift Data Foundation services including the backend Red Hat Ceph Storage (RHCS) cluster.
- For External mode deployments, upgrading OpenShift Data Foundation only upgrades the OpenShift Data Foundation service while the backend Ceph storage cluster remains untouched and needs to be upgraded separately.
  Hence, we recommend upgrading RHCS along with OpenShift Data Foundation in order to get new feature support, security fixes, and other bug fixes. Since we do not have a strong dependency on RHCS upgrade, you can upgrade the OpenShift Data Foundation operator first followed by RHCS upgrade or vice-versa. See solution to know more about RHCS releases.

When a new z-stream release becomes available, the upgrade process triggers automatically if the update strategy was set to **Automatic**. If the update strategy is set to **Manual** then use the following procedure.

Prerequisites

- Ensure that the **OpenShift Container Platform** cluster has been updated to the latest stable release of version 4.22.x, see Updating Clusters.
- Ensure that the **OpenShift Data Foundation** cluster is healthy and data is resilient.
  - Navigate to **Storage → Data Foundation → Storage Systems** tab and then click on the storage system name.
  - Check for the green tick on the status card of **Overview - Block and File** and **Object** tabs. Green tick indicates that the storage cluster, object service and data resiliency is healthy.
- Ensure that all **OpenShift Data Foundation** Pods, including the operator pods, are in **Running** state in the **openshift-storage** namespace.
  To view the state of the pods, on the OpenShift Web Console, click **Workloads → Pods**. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

- Ensure that you have sufficient time to complete the OpenShift Data Foundation update process, as the update time varies depending on the number of OSDs that run in the cluster.

Procedure

1. On the OpenShift Web Console, navigate to **Ecosystem → Installed Operators**.
2. Select **openshift-storage** project.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

3. Click the **OpenShift Data Foundation** operator name.
4. Click the **Subscription** tab.
5. If the **Upgrade Status** shows **require approval**, click on **requires approval** link.
6. On the **InstallPlan Details** page, click **Preview Install Plan**.
7. Review the install plan and click **Approve**.
8. Wait for the Status to change from **Unknown** to **Created**.
9. After the operator is successfully installed, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.

Verification

- Check the **Version** after the OpenShift Data Foundation name and check the operator status.
  - Navigate to **Ecosystem → Installed Operators** and select the **openshift-storage** project.
  - When the upgrade completes, the version updates to a new version number for OpenShift Data Foundation and status changes to **Succeeded** with a green tick.
- Verify that the OpenShift Data Foundation cluster is healthy and data is resilient.
  - Navigate to **Storage → Data Foundation → Storage Systems** tab and then click on the storage system name.
  - Check for the green tick on the status card of **Overview - Block and File** and **Object** tabs. Green tick indicates that the storage cluster, object service and data resiliency is healthy
- If verification steps fail, contact Red Hat Support.

CHAPTER 5. CHANGING THE UPDATE APPROVAL STRATEGY

To ensure that the storage system gets updated automatically when a new update is available in the same channel, we recommend keeping the update approval strategy to **Automatic**. Changing the update approval strategy to **Manual** will need manual approval for each upgrade.

Procedure

1. Navigate to **Ecosystem → Installed Operators**.
2. Select **openshift-storage** from the **Project** drop-down list.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

3. Click on **OpenShift Data Foundation** operator name
4. Go to the **Subscription** tab.
5. Click on the **pencil** icon for changing the **Update approval**.
6. Select the update approval strategy and click **Save**.

Verification

- Verify that the Update approval shows the newly selected approval strategy after it.

CHAPTER 6. UPDATING THE OPENSHIFT DATA FOUNDATION EXTERNAL SECRET

After upgrading to the latest version of OpenShift Data Foundation, update the OpenShift Data Foundation external secret.



NOTE

Updating the external secret is not required for batch updates. For example, when updating from OpenShift Data Foundation 4.22.x to 4.22.y.

Prerequisites

- Update the OpenShift Container Platform cluster to the latest stable release of 4.22.z, see Updating Clusters.
- Ensure that the OpenShift Data Foundation cluster is healthy and the data is resilient. Navigate to **Storage → Data Foundation → Storage Systems** tab and then click on the storage system name.
  - On the **Overview** - **Block and File** tab, check the Status card and confirm that the *Storage cluster* has a green tick indicating it is healthy.
  - Click the **Object** tab and confirm *Object Service* and *Data resiliency* has a green tick indicating it is healthy. The RADOS Object Gateway is only listed in case RADOS Object Gateway endpoint details are included while deploying OpenShift Data Foundation in external mode.
- Red Hat Ceph Storage must have a Ceph dashboard installed and configured.

Procedure

1. Download the OpenShift Data Foundation version of the **ceph-external-cluster-details-exporter.py** python script:

```
# oc get cm rook-ceph-external-cluster-script-config -n openshift-storage -o
jsonpath='{.data.script}' | base64 --decode > ceph-external-cluster-details-exporter.py
```

2. Update permission caps on the external Red Hat Ceph Storage cluster by running **ceph-external-cluster-details-exporter.py** on any client node in the external Red Hat Ceph Storage cluster. You may need to ask your Red Hat Ceph Storage administrator to do this.

```
# python3 ceph-external-cluster-details-exporter.py --upgrade
```

The updated permissions for the user are set as:

```
client.csi-cephfs-node
key: AQCYz0piYgu/IRAAipji4C8+Lfymu9vOrox3zQ==
caps: [mds] allow rw
caps: [mgr] allow rw
caps: [mon] allow r, allow command 'osd blocklist'
caps: [osd] allow rw tag cephfs =
client.csi-cephfs-provisioner
key: AQCYz0piDUMSIxAARuGUyhLXFO9u4zQeRG65pQ==
```

```
caps: [mgr] allow rw
caps: [mon] allow r, allow command 'osd blocklist'
caps: [osd] allow rw tag cephfs metadata=*
client.csi-rbd-node
key: AQCYz0pi88IKHhAAvzRN4fD90nkb082ldrTaHA==
caps: [mon] profile rbd, allow command 'osd blocklist'
caps: [osd] profile rbd
client.csi-rbd-provisioner
key: AQCYz0pi6W8IIBAAgRJfrAW7kZfucNdqJqS9dQ==
caps: [mgr] allow rw
caps: [mon] profile rbd, allow command 'osd blocklist'
caps: [osd] profile rbd
```

3. Run the previously downloaded python script using one of the following options based on the method you used during deployment, either a configuration file or command-line flags.

   a. **Configuration file**
   Create a **config.ini** file that includes all of the parameters used during initial deployment. Run the following command to get the configmap output which contains those parameters:

   ```
   $ oc get configmap  -namespace openshift-storage external-cluster-user-command --output jsonpath='{.data.args}'
   ```

   Add the parameters from the previous output to the **config.ini** file. You can add additional parameters to the **config.ini** file to those used during deployment. See b for descriptions of the parameters.

   Example **config.ini** file:

   ```
   [Configurations]
   format = bash
   cephfs-filesystem-name = <filesystem-name>
   rbd-data-pool-name = <pool_name>
   ...
   ```

   Run the python script:

   ```
   # python3 ceph-external-cluster-details-exporter.py --config-file <config-file>
   ```

   Replace **<config-file>** with the path to the **config.ini** file.

   b. **Command-line flags**
   Run the previously downloaded python script and pass the parameters for your deployment. Make sure to use all the flags that you used in the original deployment including any optional argument that you have used. You can also add additional flags to those used during deployment. See b for descriptions of the parameters.

   ```
   # python3 ceph-external-cluster-details-exporter.py --rbd-data-pool-name _<rbd block pool name>_ --monitoring-endpoint _<ceph mgr prometheus exporter endpoint>_ --monitoring-endpoint-port _<ceph mgr prometheus exporter port>_ --rgw-endpoint _<rgw endpoint>_ --run-as-user _<ocs_client_name>_  [optional arguments]
   ```

   Mandatory and optional parameters used during upgrade

Parameter	Description
rbd-data-pool-name	(Mandatory) Used for providing block storage in OpenShift Data Foundation.
rgw-endpoint	(Optional) Provide this parameter if object storage is to be provisioned through Ceph RADOS Gateway for OpenShift Data Foundation. Provide the endpoint in the following format: **<ip_address>:<port>**.
monitoring-endpoint	(Optional) Accepts comma separated list of IP addresses of active and standby mgrs reachable from the OpenShift Container Platform cluster. If not provided, the value is automatically populated.
monitoring-endpoint-port	(Optional) The port associated with the ceph-mgr Prometheus exporter specified by **--monitoring-endpoint**. If not provided, the value is automatically populated.
run-as-user	(Mandatory) The client name used during OpenShift Data Foundation cluster deployment. Use the default client name **client.healthchecker** if a different client name was not set.
rgw-pool-prefix	(Optional) The prefix of the RGW pools. If not specified, the default prefix is **default**.
rgw-tls-cert-path	(Optional) The file path of the RADOS Gateway endpoint TLS certificate.
rgw-skip-tls	(Optional) This parameter ignores the TLS certification validation when a self-signed certificate is provided (NOT RECOMMENDED).
ceph-conf	(Optional) The name of the Ceph configuration file.
k8s-cluster-name	(Optional) Kubernetes cluster name.
cluster-name	(Optional) The Ceph cluster name.
output	(Optional) The file where the output is required to be stored.
cephfs-metadata-pool-name	(Optional) The name of the CephFS metadata pool.
cephfs-data-pool-name	(Optional) The name of the CephFS data pool.
cephfs-filesystem-name	(Optional) The name of the CephFS filesystem.

rbd-metadata-ec-pool-name	(Optional) The name of the erasure coded RBD metadata pool.
dry-run	(Optional) This parameter helps to print the executed commands without running them.

1. Save the JSON output generated after running the script in the previous step.
Example output:

[{"name": "rook-ceph-mon-endpoints", "kind": "ConfigMap", "data": {"data": "xxx.xxx.xxx.xxx:xxxx", "maxMonId": "0", "mapping": "{}"}}, {"name": "rook-ceph-mon", "kind": "Secret", "data": {"admin-secret": "admin-secret", "fsid": "<fs-id>", "mon-secret": "mon-secret"}}, {"name": "rook-ceph-operator-creds", "kind": "Secret", "data": {"userID": "<user-id>", "userKey": "<user-key>"}}, {"name": "rook-csi-rbd-node", "kind": "Secret", "data": {"userID": "csi-rbd-node", "userKey": "<user-key>"}}, {"name": "ceph-rbd", "kind": "StorageClass", "data": {"pool": "<pool>"}}, {"name": "monitoring-endpoint", "kind": "CephCluster", "data": {"MonitoringEndpoint": "xxx.xxx.xxx.xxxx", "MonitoringPort": "xxxx"}}, {"name": "rook-ceph-dashboard-link", "kind": "Secret", "data": {"userID": "ceph-dashboard-link", "userKey": "<user-key>"}}, {"name": "rook-csi-rbd-provisioner", "kind": "Secret", "data": {"userID": "csi-rbd-provisioner", "userKey": "<user-key>"}}, {"name": "rook-csi-cephfs-provisioner", "kind": "Secret", "data": {"adminID": "csi-cephfs-provisioner", "adminKey": "<admin-key>"}}, {"name": "rook-csi-cephfs-node", "kind": "Secret", "data": {"adminID": "csi-cephfs-node", "adminKey": "<admin-key>"}}, {"name": "cephfs", "kind": "StorageClass", "data": {"fsName": "cephfs", "pool": "cephfs_data"}}, {"name": "ceph-rgw", "kind": "StorageClass", "data": {"endpoint": "xxx.xxx.xxx.xxxx", "poolPrefix": "default"}}, {"name": "rgw-admin-ops-user", "kind": "Secret", "data": {"accessKey": "<access-key>", "secretKey": "<secret-key>"}}]

2. Upload the generated JSON file.
   a. Log in to the OpenShift Web Console.
   b. Click **Workloads → Secrets**
   c. Set project to **openshift-storage**.
   d. Click **rook-ceph-external-cluster-details**.
   e. Click **Actions (⋮) → Edit Secret**
   f. Click **Browse** and upload the JSON file.
   g. Click **Save**.

Verification

- To verify that the OpenShift Data Foundation cluster is healthy and data is resilient, navigate to **Storage → Data Foundation → Storage Systems** tab and then click on the storage system name.
  - On the **Overview → Block and File** tab, check the Details card to verify that the RHCS dashboard link is available and also check the Status card to confirm that the *Storage Cluster* has a green tick indicating it is healthy.
  - Click the **Object** tab and confirm *Object Service* and *Data resiliency* has a green tick

indicating it is healthy. The RADOS Object Gateway is only listed in case RADOS Object Gateway endpoint details are included while deploying OpenShift Data Foundation in external mode.

- If verification steps fail, contact Red Hat Support .